Computational Dynamics Q – Assignment Q3

Table of Contents

Q3 – FORWARD DYNAMICS

Using the same setup as in Assignment Q2, we will now investigate how the space shuttle moves relative to an inertial observer when no vectored thrust is applied to counteract the motion of the space shuttle due to a similar movement of the payload compared to Q2.

This will be done using forward dynamics, i.e. we prescribe the generalized forces in the hinges of the robotic arm and the space shuttle to determine the corresponding motion of the system.

To allow free motion between the inertial observer and the space shuttle a new hinge, $H(5)$, needs to be introduced between the two, where $H(5) = I_6$ with I_6 being the 6×6 identity matrix. Figure 1 shows the corresponding modified body diagram.

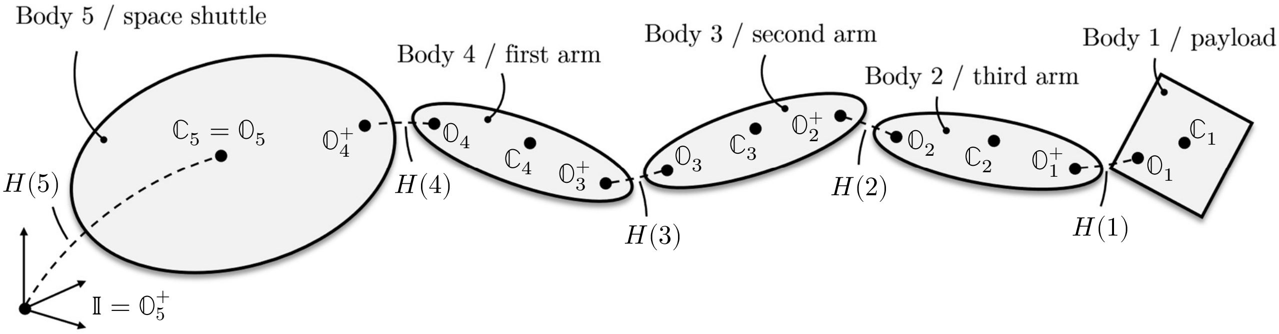


Figure 1

The base body is now the left-most frame, $\mathbb{I}$, meaning that the number of movable bodies is now incremented to $n = 5$. Note that the equal signs in Figure 1 simply means that the corresponding frames are coincident (but not necessarily aligned). The properties of the bodies are the same as in Assignment Q2 with the introduction of the space shuttle mass, $m_5 = 294.000$ kg and space shuttle 2nd order moment of inertia, ${}^{\mathbb{C}_5}\mathscr{J}_5 = \mathrm{diag}(3.870, 29.04, 30.30) \cdot 10^6\ \mathrm{kgm}^2$.

We define the generalized velocities for the kth body, $\beta(k) \in \mathbb{R}^{r_v(k)}$, and the generalized acceleration for the kth body, $\gamma(k) \in \mathbb{R}^{r_v(k)}$, the same way as in Assignment Q2 with the addition of $\beta(5) \triangleq \mathcal{V}(5)$. We also denote the generalized forces for the kth body as $\mathcal{T}(k) \in \mathbb{R}^{r_v(k)}$.

Because $H(5)$ contains prismatic components, unlike all the other hinges, we now need to deal with the effect of translation in the hinges. For this reason, we extend the definition of $\theta(k)$ and renaming it as the generalized coordinates vector for the kth body by defining it as,

$$\theta(k) \triangleq [\sigma^\top(k) \quad \wp^\top(k)]^\top \in \mathbb{R}^{r_p(k)} \ ,$$

with $\sigma(k)$ being the previously defined generalized orientations vector, i.e. for Assignment Q2 we had $\sigma(k) = \theta(k)$, with the addition that $\sigma(5) \triangleq {}^6\underline{q}_5$ and $\wp(k)$ being the generalized translations vector for the kth body which is only defined for $k = 5$ as $\wp(5) \triangleq l(\mathbb{O}_5^+, \mathbb{O}_5) \in \mathbb{R}^3$.

We also introduce the spatial coordinates vector for the kth body, $\mathcal{X}(k)$, defined as,

$$\mathcal{X}(k) \triangleq [{}^{k+1}\underline{q}_k^\top \quad l^\top(\mathbb{O}_k, \mathbb{O}_{k-1})]^\top \in \mathbb{R}^7 \ .$$

Q3.1 ATBI algorithm

Define a function that uses the $\mathcal{O}(n)$ ATBI forward dynamics algorithm by mapping the stacked generalized vectors, $\theta \in \mathbb{R}^{17}$, $\beta \in \mathbb{R}^{14}$ and $\mathcal{T} \in \mathbb{R}^{14}$, with the stacked spatial vectors, $\mathcal{X} \in \mathbb{R}^{35}$, $\mathcal{V} \in \mathbb{R}^{30}$ and $\alpha \in \mathbb{R}^{30}$.

Hints:

- For the first and second scatter sweep, copy the scatter algorithm from the previous assignment but exclude the calculation of the spatial acceleration since this is not given and modify the calculation of the rigid body transformation matrix to account for hinges with prismatic components since the assumption of $\phi(\mathbb{O}_{k+1}, \mathbb{O}_k) = \phi(\mathbb{O}_{k+1}, \mathbb{O}_k^+)$ is no longer valid. Keep the calculation for the Coriolis acceleration but ensure that it accounts for hinges with prismatic components.
- If you want the spatial forces as well, you can copy the gather loop from previous assignment to the end of the ATBI algorithm or use $\mathfrak{f}(k) = \mathcal{P}(k)\big(\alpha(k) - \mathfrak{a}(k)\big) + \mathfrak{z}(k)$.

Q3.2 Equation of motion

Define the equation of motion that maps the state vector, $S = [\theta^\top \quad \beta^\top]^\top$, to the derivative of the state vector, $\dot{S} = [\dot{\theta}^\top \quad \dot{\beta}^\top]^\top$. Use the ABTI algorithm to determine $\dot{\beta}$ with the generalized forces $\mathcal{T}(1)$, $\mathcal{T}(2)$, $\mathcal{T}(3)$, $\mathcal{T}(4)$ interpolated from the attached `generalized_forces.csv` file and,

$$\mathcal{T}(5) = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^\top \ .$$

Note that the first column in the file is the time.

Hints:

- As in Assignment Q1 ensure that the rotation of $\mathbb{O}_5$ is accounted for when determining $\dot{\wp}(5)$ which can require a different approach depending on which frame you choose to represent $\wp(5)$ in.
- A linear interpolation of the file should be sufficient.

Q3.3 Simulation

Simulate the system for $1{,}000$ sec using the previously defined functions and the same initial conditions as in Assignment Q2 with the addition of,

$$\wp|_{t=0} = [0 \quad 0 \quad 0]^\top \ ,$$

$$\sigma(5)|_{t=0} = [0 \quad 0 \quad 0 \quad 1]^{\top} \ .$$

Plot the spatial accelerations for the space shuttle and the payload in their respective $\mathbb{O}$-frames – see Figure 3.

Hints:

- The generalized forces file is designed such that the motion resembles that of Assignment Q2 with the end velocity being close to zero.
- To troubleshoot properly, consider modifying the code from Assignment Q2 such that it outputs the generalized forces for all bodies including body 5, which can then be used as an input for this simulation. The expected motion should then match between the assignments. Note that the generalized forces retrieved from the previous assignment are different from the forces in the attached file even though the motions are similar since the motion for this assignment now also must account for the movement of the space shuttle.
- The expected motion should resemble that of Figure 2 (optional plot) and the spatial accelerations of the space shuttle and the payload should resemble that of Figure 3.

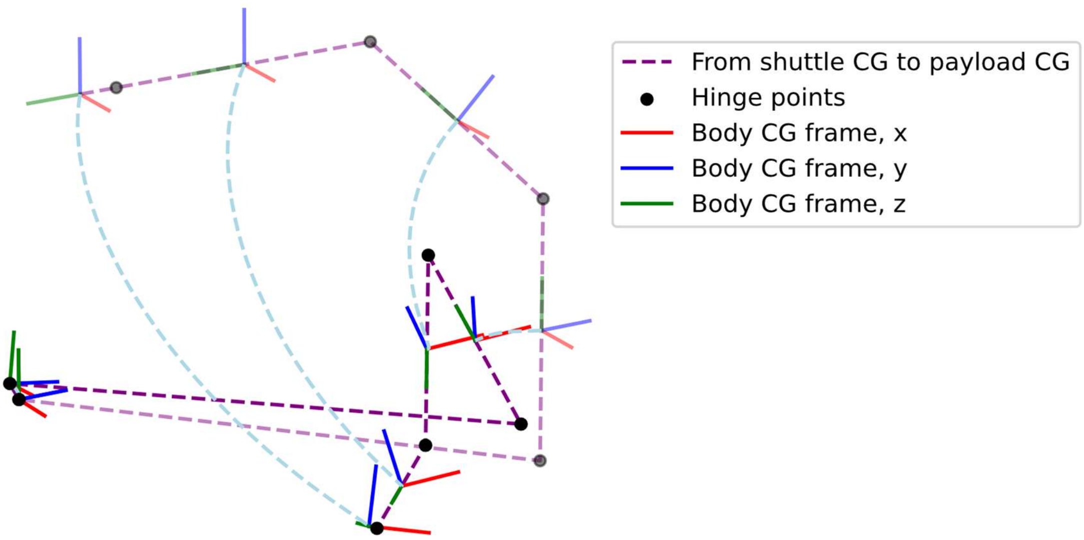
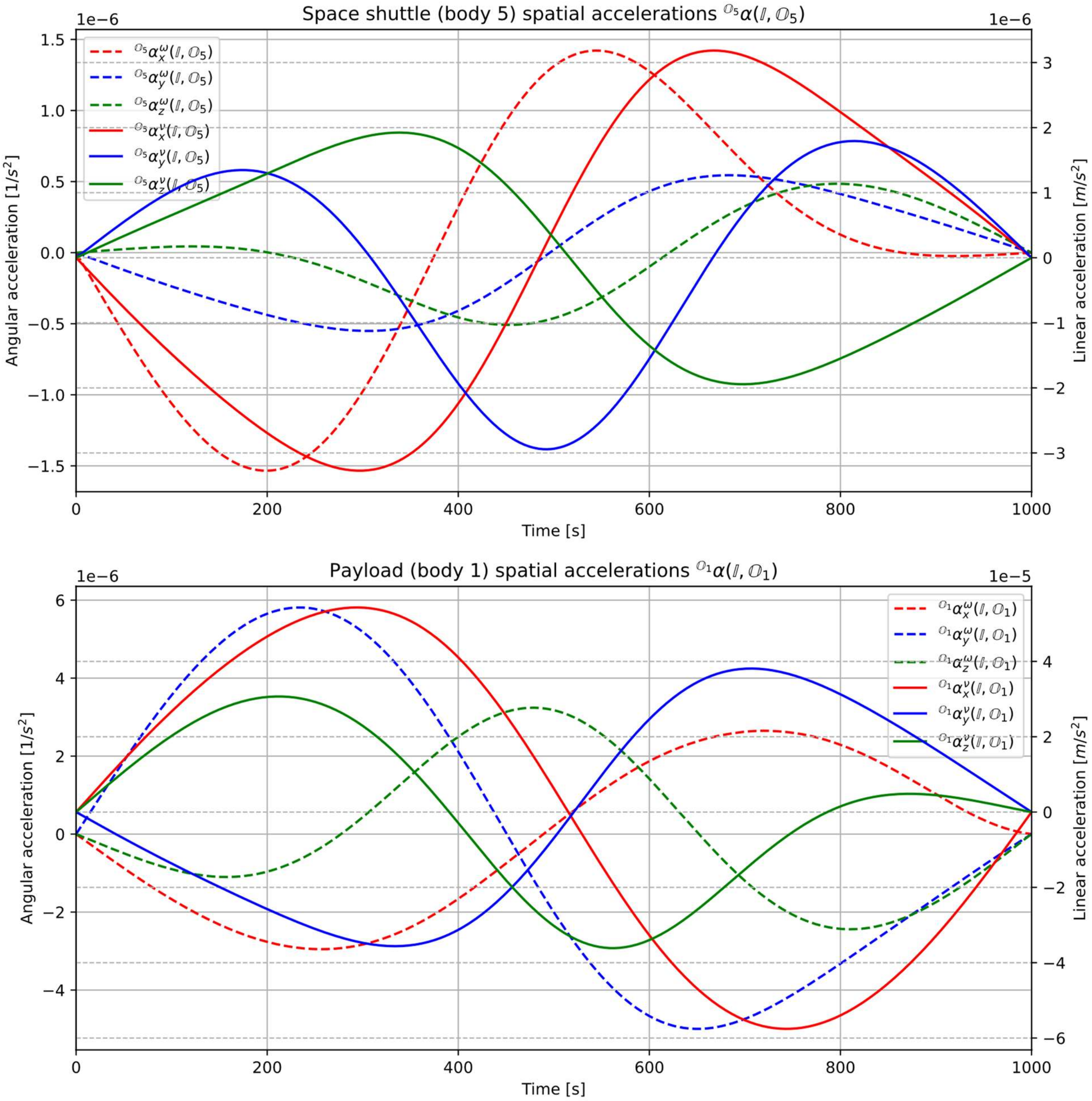


Figure 2



Space shuttle (body 5) spatial accelerations ${}^{\mathbb{O}_5}\alpha(\mathbb{I},\mathbb{O}_5)$
1e−6
1e−6
${}^{\mathbb{O}_5}\alpha^{\omega}_x(\mathbb{I},\mathbb{O}_5)$
${}^{\mathbb{O}_5}\alpha^{\omega}_y(\mathbb{I},\mathbb{O}_5)$
${}^{\mathbb{O}_5}\alpha^{\omega}_z(\mathbb{I},\mathbb{O}_5)$
${}^{\mathbb{O}_5}\alpha^{v}_x(\mathbb{I},\mathbb{O}_5)$
${}^{\mathbb{O}_5}\alpha^{v}_y(\mathbb{I},\mathbb{O}_5)$
${}^{\mathbb{O}_5}\alpha^{v}_z(\mathbb{I},\mathbb{O}_5)$
Angular acceleration $[1/s^2]$
Linear acceleration $[m/s^2]$
Time [s]
Payload (body 1) spatial accelerations ${}^{\mathbb{O}_1}\alpha(\mathbb{I},\mathbb{O}_1)$
1e−6
1e−5
${}^{\mathbb{O}_1}\alpha^{\omega}_x(\mathbb{I},\mathbb{O}_1)$
${}^{\mathbb{O}_1}\alpha^{\omega}_y(\mathbb{I},\mathbb{O}_1)$
${}^{\mathbb{O}_1}\alpha^{\omega}_z(\mathbb{I},\mathbb{O}_1)$
${}^{\mathbb{O}_1}\alpha^{v}_x(\mathbb{I},\mathbb{O}_1)$
${}^{\mathbb{O}_1}\alpha^{v}_y(\mathbb{I},\mathbb{O}_1)$
${}^{\mathbb{O}_1}\alpha^{v}_z(\mathbb{I},\mathbb{O}_1)$
Angular acceleration $[1/s^2]$
Linear acceleration $[m/s^2]$
Time [s]

Figure 3